

# XRPL Validator — Daily Engineering Report • 2026-04-16

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Compiled retrospectively on 2026-09-23 from the git history. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the XRP Ledger itself (amendments) and XRPL research.

## Summary

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- 7 commits (xrpl-validator 7)

## Engineering changes

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- **fix(sync): skip retry+full-scan on mismatch, just log and move on** ( a779791 , xrpl-validator) The 3x retry + 5min full scan never healed a real divergence - it only helped with transient rippled RPC staleness. In practice it blocked sync for 5+ minutes after every mismatch, preventing any further mismatches from being captured for diagnosis.
- **fix(sync): update hasher during fast-path gap catchup** ( 5957491 , xrpl-validator) ROOT CAUSE FOUND: When ws\_sync falls behind (network closed multiple ledgers since our last sync), it processes gap ledgers via "fast path" which writes to state.rocks but skips the hash check. The fast path updated leaf\_cache (used only for disk persistence) but did NOT update the live hasher.
- **perf(sync): hash-check every ledger + add per-stage timing** ( 4f2122c , xrpl-validator) Fix #1: force compute\_hash=true on every ledger. Previously only the last ledger in a gap-catchup batch got hash-verified, so the validation match count was capped by the "catch up to tip" rate. Now every ledger applied is a ledger verified.
- **fix(sync): fetch account\_hash for every ledger, not just batch tail** ( 6cc3fb6 , xrpl-validator) compute\_hash was always true after the last fix, but account\_hash was still only populated when process\_seq == closed\_seq. Gap ledgers computed their hash and had nothing to compare against — so no MATCH/MISMATCH log fired and shadow hash stayed at 0.

## Docs and operations

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- **diag(sync): always snapshot hasher so overlay\_diff dumps every mismatch** ( e80aaad , xrpl-validator)
- **diag: add check\_key tool to look up SLE in state.rocks read-only** ( 49553ad , xrpl-validator)
- **diag(ffl): log sender AccountRoot sequence at terPRE\_SEQ + tesSUCCESS** ( 9711baf , xrpl-validator)

## Appendix — commits

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| Time  | Repo           | Commit  | Subject                                                                 |
|-------|----------------|---------|-------------------------------------------------------------------------|
| 06:25 | xrpl-validator | a779791 | fix(sync): skip retry+full-scan on mismatch, just log and move on       |
| 07:04 | xrpl-validator | e80aaad | diag(sync): always snapshot hasher so overlay_diff dumps every mismatch |
| 08:26 | xrpl-validator | 49553ad | diag: add check_key tool to look up SLE in state.rocks read-only        |

| Time  | Repo           | Commit  | Subject                                                               |
|-------|----------------|---------|-----------------------------------------------------------------------|
| 09:13 | xrpl-validator | 9711baf | diag(ffl): log sender AccountRoot sequence at terPRE_SEQ + tesSUCCESS |
| 11:17 | xrpl-validator | 5957491 | fix(sync): update hasher during fast-path gap catchup                 |
| 20:41 | xrpl-validator | 4f2122c | perf(sync): hash-check every ledger + add per-stage timing            |
| 20:59 | xrpl-validator | 6cc3fb6 | fix(sync): fetch account_hash for every ledger, not just batch tail   |